Final Exam

● Graded

Student

Bilol Abdilxayev

Total Points

41.5 / 45 pts

Question 1

Overall Grades **41.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 3.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Bilol Abdixayev	ba211	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

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- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE:

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	5	1.5	5	41.5

Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

Coefficient Matrix

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 1 & 0 & 2 \end{bmatrix}$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_3 \leftrightarrow R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right] \quad R_3 \leftarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right] \quad R_2 \leftarrow -\frac{1}{2} R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -3 & -4 & -1 \end{array}\right] \quad R_3 \leftarrow R_3 + 3R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{5}{2} \end{array}\right]$$

$$R_3 \leftarrow 2R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -5 \end{array}\right] \quad R_2 \leftarrow R_2 - \frac{3}{2} R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$$R_1 \leftarrow R_1 - 2R_3$$

Answer $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$ no free parameters

exactly one solution.

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

① RREF $(A|_{5})$

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{matrix} R_1 \leftarrow \frac{1}{2} R_1 \\ R_2 \leftarrow \frac{1}{5} R_2 \end{matrix} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{matrix} R_2 \leftarrow R_2 + R_1 \\ R_4 \leftarrow R_4 - R_1 \end{matrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} R_2 \leftarrow \frac{1}{2} R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{matrix} R_3 \leftarrow R_3 - R_2 \\ R_4 \leftarrow R_4 - R_2 \end{matrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} R_1 \leftarrow R_1 - 2R_2 \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Since

$$RREF(A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

①

$col(A) = span\{v_1, v_2\}$ where v_1 and v_2 are col vectors.

basis of a column space of A is $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

②

$row(A) = span\{u_1, u_2\}$ where u_1 and u_2 are row vectors.

So basis of a row(A) is $\left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}^T, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix}^T \right\}$

③ $null(A) = \{x \in R^3 : Ax = 0\}$, so

basis of col(A) is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$$A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

let $B = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$, assum B is invertible (otherwise we do not have enough info to find A), then

$$AB = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

$$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}B^{-1}$$

B^{-1}:

$$\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right]$$

① $R_2 \leftrightarrow R_1$

$$\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&-1&1&1&1\end{array}\right]$$

$R_2 \leftarrow R_2 + R_1$

$R_3 \leftarrow R_3 + R_2$

$R_4 \leftarrow (-1)R_4$

$$\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right]$$

B^{-1}

So $B^{-1} = B$

$$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

row-wise multiplication

$0\ 1\ 0$

$+$

$2(-1\ -1\ -1) \rightarrow [-2\ -1\ -2]$

$0\ 1\ 0$

$+$

$1\ 0\ 0 \rightarrow [0\ -1\ -2]$

$+$

$-2\ -2\ -2$

So $A = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$

5/5

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$\det(A-\lambda I)=0$

1) Find eigenvalues

By cofactor expansion choose third column to compute derivative

$$\det\left(\begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}\right)=0 \Rightarrow (1-\lambda)\det\left(\begin{bmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{bmatrix}\right)=0$$

↳ upper triangular matrix

$$\Rightarrow (2-\lambda)(1-\lambda)(2-\lambda)=0$$

so eigenvalues are 1, 1, 2, or 1 with algebraic multiplicity 2 and 2 with algebraic multiplicity 1

2) Find eigenspace of 1 (E_1)

red eigenvalue 1

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \begin{matrix} R_1 \leftarrow R_1 - 2R_2 \\ R_3 \leftarrow -\frac{1}{2}R_3 \end{matrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} R_3 \leftrightarrow R_1$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} R_1 \leftarrow R_1 - R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ so } \text{Null}(A-I) = \text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$$

Therefore, geometric multiplicity of eigenvalue 1 is 1, meaning not enough vectors to diagonalize A.

Eigenspace of 2: $E_2 \Rightarrow \text{Null}(A-2I)$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} R_3 \leftarrow R_3 - 2R_1 \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & -6 & -1 \end{bmatrix} R_2 \leftarrow (-1)R_2 \; R_3 \leftarrow \left(-\frac{1}{6}\right)R_3 \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & \frac{1}{6} \end{bmatrix} R_2 \leftarrow R_3 \; R_1 \leftarrow R_1 + 2R_3$$

$$\begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{6} \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow E_2 = \text{span}\left\{\begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}\right\} \text{ choose eigenvector } \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$$

3) find V_2 for eigenspace E_1

$$(A-\lambda I)V_2 = V_1 \Rightarrow \text{solve system } \left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \Rightarrow V_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$$

Please check back of the page

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

let $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$ so $f(x,y,z) = \langle x,y,z \rangle A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\det\left(\begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}\right) = (3-\lambda)\det\begin{pmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{pmatrix} + \det\left(\begin{bmatrix} 0 & 1 \\ 3-\lambda & -1 \end{bmatrix}\right) = (3-\lambda)\big((3-\lambda)(4-\lambda)-1\big) + (-1)(3-\lambda) =$$

$$= (3-\lambda)\left(\lambda^2 - 7\lambda + 12 - 2\right) = (3-\lambda)(\lambda-5)(\lambda-2)$$

& eigenvalues are 3, 5, and 2

$$E_2 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \quad R_3 \leftarrow R_3 - R_1 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \quad R_3 \leftarrow R_3 + R_2 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \; V_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$E_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad R_3 \leftrightarrow R_1 \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R_3 \leftarrow R_3 + R_2 \\ R_2 \leftarrow R_2 + R_3 \end{matrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; V_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$E_5 = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \quad \begin{matrix} R_1 \leftarrow R_1 + 2R_3 \\ R_1 \leftrightarrow R_3 \end{matrix} \begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ 0 & -2 & -1 \end{bmatrix} \quad \begin{matrix} R_3 \leftarrow R_3 - R_2 \\ R_2 \leftarrow -\frac{1}{2} R_2 \\ R_1 \leftarrow R_1 + R_2 \end{matrix} \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}, V_5 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Make unit vectors

$$V_2 \Rightarrow \frac{1}{\|V_2\|} V_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$V_3 \Rightarrow \frac{1}{\|V_3\|} V_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$V_5 \Rightarrow \frac{1}{\|V_5\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

let $Q = \begin{bmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$ $\quad Q^{-1} = Q^T$

$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ $\quad f(x,y,z) = \langle x,y,z \rangle Q D Q^T \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

check back page please

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

Solve system

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \quad R_2 \leftarrow R_2 + R_3 \quad \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \\ -2 & 2 & 0 & -8 \end{array}\right] \quad \begin{array}{l} R_2 \leftarrow -\frac{1}{2} R_2 \\ R_2 \leftrightarrow R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right] \quad R_3 \leftarrow R_3 - 2R_2 \quad \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right] \quad \begin{array}{l} R_3 \leftarrow -\frac{1}{3} R_3 \\ R_1 \leftarrow R_1 + R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right] \quad \begin{array}{l} R_1 \leftarrow R_1 - R_3 \\ R_2 \leftarrow R_1 - R_2 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Answer:

$a = 3$

$b = -1$

$c = 1$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Use Gram Schmidt Orthogonalization

Let $v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

Choose $u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, then orthogonal $u_2 = v_2 - \frac{u_1 \cdot v_2}{u_1 \cdot u_1} u_1$

$u_1 \cdot u_1 = 1 \cdot 1 + 1 \cdot 1 = 2$

$u_1 \cdot v_2 = 3 \cdot 0 + 1 \cdot 0 + 1 \cdot 2 + 0 \cdot (-1) = 2$

So $u_2 = v_2 - \frac{2}{2} u_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

Now, normalize

$$\frac{u_2}{\|u_2\|} = \begin{bmatrix}\frac{3}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\\\frac{1}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\end{bmatrix} \qquad \frac{u_1}{\|u_1\|} = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}$$

Answer: $V = span\left\{\begin{bmatrix}\frac{3}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\\\frac{1}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\end{bmatrix}, \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}\right\}$ orthonormal basis

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Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$A^T = -A$ are skew-symmetric family of matrices, the diagonal is forced to be 0

Assume general matrix $A \in \mathbb{R}^{n \times n}$: $A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & & \\ \vdots & & & & \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$

① Suppose some a_{ii} $1 \le i \le n$ is not zero, then i^{th} entry of A^T at (i,i) is same which contradicts statement $A^T = -A$ (entry wise), also for every other entry $a_{ij} = -a_{ji}$

② Use property that $\det A = \det A^T$ (could be proven using algebraic derivation)

③ Notice that any for any odd n you can choose vector $v = \langle a_{n,1}, a_{n-1,1}, a_{n-2,1} \cdots a_{2,1} \rangle^T$ such that it becomes an eigen vector for eigenvalue "0"

To show $\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & & \\ \vdots & & & \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} \begin{bmatrix} a_{n,1} \\ a_{n-1,1} \\ a_{n-2,1} \\ \vdots \\ a_{2,1} \end{bmatrix} = 0$

$A = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & -2 \\ -3 & 2 & 0 \end{bmatrix}$, $n=3$

$A \begin{bmatrix} -3 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 7 \end{bmatrix} \neq \vec{0}$

Dot row $(a_{n-1,1})(a_{1,2}) - (a_{n-2,1})(a_{1,3})$ ... $= (a_{1,1})(a_{1,n})$ → we have $(a_{2,1}) \cdot (a_{2,n-2})$ → already 0, since $(a_{i,1}) > 0$

3/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

(1)

Square both sides $\|\vec{u}\|^2 = \vec{u}\cdot\vec{u}$

$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$ use property that $\vec{v}\cdot\vec{v} = \|\vec{v}\|^2$

$(\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v}) = \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$ (Associativity of dot products and addition of vectors)

$(\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v}) = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$

Given

$\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 \Rightarrow$

$2\vec{u}\cdot\vec{v} = -2\vec{u}\cdot\vec{v}$

$(2+2)\ \vec{u}\cdot\vec{v} = 0 \Rightarrow \vec{u}\cdot\vec{v} = 0$

which is the definition of orthogonality.